Sentiment Analysis of Movie Reviews: A Comparative Study of Classic and Transformer-Based Models

1. Problem Statement

This project aims to classify movie review phrases into five sentiment categories (0 = very negative to 4 = very positive), replicating and extending the 2014 "Sentiment Analysis on Movie Reviews" paper by Cai & Wang. We establish classic ML baselines and compare them to modern transformer-based architectures to identify the most effective approach for fine-grained sentiment classification.

2. Methodology & Implementation

The Kaggle Rotten Tomatoes dataset was used with a 20% hold-out test set for evaluation. In Phase 1, classic models were trained using text preprocessing (lowercasing, punctuation and stopword removal, lemmatization) followed by Bag-of-Words with unigrams, bigrams, and combined n-grams. The models included Naive Bayes, Softmax Regression, and Linear SVM.

In Phase 2, transformer-based models were applied: a fine-tuned `bert-base-uncased` model with a classification head, and a hybrid BERT + BiLSTM model where BERT embeddings were passed through a BiLSTM layer to better capture sequential patterns.

All models were implemented in Python on Google Colab with GPU acceleration. Classic models used scikit-learn, NLTK, and pandas, while transformer models were built with PyTorch and Hugging Face Transformers.

3. Results & Conclusions

Our experiments revealed a clear performance hierarchy, with the deep learning models significantly outperforming the classic baselines. The BERT+BiLSTM model achieved the highest accuracy, confirming that a hybrid approach combining deep contextual embeddings with sequential modeling is the most effective for this task.

Model	Architecture	Test Accuracy
BERT + BiLSTM	Fine-Tuned Hybrid Transformer	67.98%
BERT	Fine-Tuned Transformer	67.84%
Softmax Regression	Classic ML (Combined N-Grams)	66.25%
SVM	Classic ML (Combined N-Grams)	64.78%
Naive Bayes	Classic ML (Combined N-Grams)	61.41%

Conclusions:

1. State-of-the-art transformer models provide a clear performance advantage over traditional Bag-of-Words methods.
2. Softmax Regression proved to be the strongest and most reliable classic baseline model.
3. The slight performance edge of the BERT+BiLSTM model over the standard BERT suggests that adding a sequential layer to model the output of the transformer is beneficial.

5. Challenges & Key Learnings

A major challenge was class imbalance: the "neutral" sentiment (label 2) dominated the dataset, while "very negative" (0) and "very positive" (4) were underrepresented. As a result, models showed high recall for neutral but poor performance on extreme sentiments. Addressing this would require techniques like class weighting or data augmentation, which were outside the scope of this study.